

```

a=615/18;%divisions along middle section of T-beam
b=200/7;%divisions along cross-beam

%coordinates of locations 1-18 (middle section)
x_m=a*(1:18);
y_m=0*(1:18);

%coordinates of locations 19-32, with 18 in the middle (cross-beam)
x_c=18*a*ones(15,1);
y_c=b*(-7:7);

%beam coordinates
x=[0,0,582,582,645,645,582,582];
y=[34,-34,-34,-200,-200,200,200,34];
z=[0,0,0,0,0,0,0,0];

%combine vectors of frequencies at each location together
frequency=[frequency1,frequency2,frequency3,frequency4,frequency5,frequency6...,
    frequency7,frequency8,frequency9,frequency10,frequency11,frequency12...,
    frequency13,frequency14,frequency15,frequency16,frequency17,frequency18...,
    frequency19,frequency20,frequency21,frequency22,frequency23,frequency24...,
    frequency25,frequency26,frequency27,frequency28,frequency29,frequency30...,
    frequency31,frequency32];

%combine vectors of imaginary components at each location together
imaginary=[imaginary1,imaginary2,imaginary3,imaginary4,imaginary5,imaginary6...,
    imaginary7,imaginary8,imaginary9,imaginary10,imaginary11,imaginary12...,
    imaginary13,imaginary14,imaginary15,imaginary16,imaginary17,imaginary18...,
    imaginary19,imaginary20,imaginary21,imaginary22,imaginary23,imaginary24...,
    imaginary25,imaginary26,imaginary27,imaginary28,imaginary29,imaginary30...,
    imaginary31,imaginary32];

figure

%define vector of identified resonant frequencies
RF=[7.5,31.3,70,182.5,225,430];

j=1;
%cycle through each resonant frequency to plot its corresponding mode shape
while j<7

    %vector of imaginary components at each beam location
    ims=zeros(32,1);

    k=1;
    %cycle through each location on the beam to find its corresponding
    %imaginary component
    while k<33

        %find the position in the frequency vector of the frequency
        %closest to the current resonant frequency
        [~,idx]=min(abs(frequency(:,k)-RF(j)));

        %locate the peak imaginary value corresponding to this frequency
        [~,idx2]=max(abs(imaginary(idx-5:idx+5,k)));

        %assign the value of this peak to the vector of imaginary
        %components at each beam location
        ims(k)=imaginary(idx-6+idx2,k);

        k=k+1;

    end

    %assign the first 18 imaginary components to a vector representing the
    %middle section of the T-beam
    ims_m=ims(1:18);

    %assign remainder of imaginary components to a vector representing the
    %cross-beam
    ims_c=zeros(15,1);
    ims_c(1:7)=ims(19:25);
    ims_c(8)=ims(18);
    ims_c(9:15)=ims(26:32);

    %generate plots
    subplot(3,2,j)

    plot3(x_m,y_m,round(ims_m),'r',LineWidth=1.2)
    hold on
    plot3(x_c,y_c,round(ims_c),'r',LineWidth=1.2)
    hold on
    plot3(x_m,y_m,-round(ims_m),'r--')
    hold on
    plot3(x_c,y_c,-round(ims_c),'r--')
    hold on
    fill3(x,y,z,'b',FaceAlpha=0.5)

```

```
hold on
ylim([-350,350])
ylabel('mm',Interpreter='latex',FontWeight='bold',FontSize=14)
xlabel('mm',Interpreter='latex',FontWeight='bold',FontSize=14)
zlabel('Imaginary',Interpreter='latex',FontWeight='bold',FontSize=14)
RESFEQ=round(RF(j),1);
title(sprintf('Mode Shape at %d', RESFEQ),FontWeight='bold',FontSize=16);
grid on
grid minor
```

```
j=j+1;
```

```
end
```